

RESEARCH ARTICLE

Design of Multi-Agent-Based Energy Management System for DC Microgrids

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ABSTRACT The increasing penetration of distributed energy resources (DER) such as photovoltaic (PV) systems, wind turbines, and energy storage units has promoted the development of DC microgrids. However, the heterogeneous characteristics of these sources and varying load demands create challenges in maintaining power balance, ensuring DC bus voltage stability, and prioritizing critical loads. Conventional centralized energy management strategies may suffer from limited flexibility and scalability. This paper proposes a multi-agent-based energy management system (EMS) for a DC microgrid composed of PV, wind, synchronous generator, battery-based energy storage system (ESS), grid connection, and critical and non-critical loads. The proposed system employs four cooperative agents, namely load, grid, DER, and ESS agents to coordinate power sharing and maintain stable operation. The proposed microgrid and control framework are modeled in MATLAB/Simulink and evaluated under eight operating scenarios. Simulation results demonstrate that the proposed approach maintains DC bus voltage stability under varying conditions and ensures continuous supply to critical loads, thereby improving system stability and power quality.

INDEX TERMS Microgrids, renewable energy, renewable energy sources, energy management, multi-agent system.

I. INTRODUCTION

In the quest for energy sustainability and green energy, microgrids have emerged as an innovative solution that reshapes traditional centralized power distribution systems [1], [2]. A microgrid is a localized energy system consisting of inter-connected loads and distributed energy resources (DERs) that can operate independently of or in conjunction with the grid. Unlike traditional grids that rely on large-scale power plants and long-distance transmission lines, microgrids are characterized by their decentralized structure, allowing communities, campuses, and even individual buildings to generate, store, and manage their electricity locally [3]. Fundamentally, microgrids represent a shift towards a more resilient, efficient, and sustainable energy future. By embracing localized generation and smart grid technologies, these innovative systems enable communities to take control of their energy destiny, paving the way for a cleaner, greener tomorrow [4].

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There are three types of microgrids: AC, DC, and hybrid microgrids [5]. AC microgrids represent the traditional approach to electricity distribution, characterized by using AC voltage to transmit and distribute power. These microgrids are compatible with existing infrastructure and grid architecture, making them common in urban and industrial environments. AC microgrids can seamlessly integrate renewable energy resources such as solar and wind power. AC microgrids are sensitive to energy losses and voltage fluctuations over long transmission distances [6]. In contrast, DC microgrids are revolutionizing energy distribution by leveraging the inherent advantages of DC voltage for local energy production and consumption. DC microgrids minimize energy losses associated with AC conversion and transmission, making them ideal for remote or off-grid applications where efficiency and reliability are crucial. Using DC compatible loads and renewable energy sources such as photovoltaic (PV) modules and battery storage systems, DC microgrids offer a modern and cost-effective solution for powering telecommunications infrastructure, data centers

and residential communities. With the rise of digital technologies and the Internet of Things (IoT), DC microgrids are poised to play a key role in shaping the future of decentralized energy systems [7]. Hybrid microgrids combine both AC and DC technologies to optimize energy generation, storage, and distribution for a variety of applications and environments. Hybrid microgrids seamlessly integrate AC and DC components, offering flexibility, resilience, and efficiency across a wide range of operating conditions [8].

The combined use of many DERs leads to the design of energy management systems in microgrids. Energy management in microgrids covers a wide range of tasks, including demand response, load forecasting, renewable energy integration and grid optimization, etc. Energy management, in essence, aims to balance supply and demand, maximize efficiency, and minimize costs while ensuring grid stability and resilience [9]. In traditional centralized systems, a single organization typically oversees energy management tasks. However, the decentralized nature of microgrids requires distributed control strategies and energy management systems that can coordinate various energy sources and respond to dynamic operating conditions. Energy management systems in microgrids can be classified as classical method-based energy management systems, energy management systems based on meta-heuristic approaches, energy management systems based on artificial intelligence (AI) methods, energy management systems based on stochastic power programming approaches, energy management systems based on model predictive control and multi-agent energy management systems [10].

Table 1 summarizes the benefits and limitations of various energy management system approaches discussed in the literature, highlighting the advantages of the proposed multi-agent based method.

A complex system of autonomous agents with limited capabilities and local knowledge that can cooperate to accomplish a global objective is called a multi-agent system (MAS). This system provides an extra benefit when developing hybrid controllers for microgrids because it was developed using both conventional control methodologies and AI-based techniques [11], [12]. With the mentioned features, multi-agent-based systems seem to be a good solution for microgrid control problems. In the dynamic environment of energy distribution, effective management of resources is crucial to ensure the reliability, efficiency, and sustainability of microgrid systems. There are many studies on energy management applications in microgrids that benefit from the power of MASs, offering smart solutions to optimize energy production, storage, and consumption [13], [14], [15]. A multi factor distributed energy management system architecture is proposed, consisting of distributed generation units, energy storage resources, and a group of loads. The proposed framework coordinates the complex energy interactions among these components and provides more robust and higher performance control compared to traditional centralized energy management systems [13]. A multi-agent-based architecture

TABLE 1. Energy management system approaches.

EMS	Benefits	Limitations
Classical Methods	Simple, well-established	Lack adaptability, scalability
AI-Based Methods	Adaptive, learns from data	Requires large datasets, black-box behavior
Meta-Heuristic Approaches	Global optimization capability	Computationally intensive
Model Predictive Control	Handles constraints well	Needs accurate models
Multi-Agent Systems	Distributed, scalable, fault-tolerant	Communication overhead, complexity in design

for the microgrid energy management system is carried out by developing different scenarios considering the microgrid energy management system performance, resource fluctuations and load changes [14].

Despite the growing interest in multi agent systems for microgrid control, most existing studies focus on simplified system models or AC microgrids. Comprehensive frameworks specifically tailored for DC microgrids that integrate diverse distributed energy resources such as wind turbines, photovoltaic systems, synchronous generators, and battery energy storage systems (ESSs) remain limited. In addition, only a few studies address real time coordination challenges among multiple agents under dynamic operating conditions. These limitations highlight the need for a robust, scalable, and fault tolerant multi-agent-based energy management system for DC microgrids.

The primary technical challenge is maintaining stable DC bus voltage while dynamically balancing power supply and demand under fluctuating renewable generation and varying load profiles. Furthermore, guaranteeing uninterrupted power supply to critical loads during islanded operation, as well as handling communication delays or failures among agents, introduces additional complexity. Addressing these challenges requires an intelligent coordination framework capable of adaptive decision making and reliable distributed control.

This study proposes a multi-agent-based energy management system for DC microgrids. To evaluate the performance and practical effectiveness of the proposed framework, a DC microgrid is designed comprising a PV system, wind turbine, synchronous generator, battery-based ESS, utility grid connection, and both critical and non-critical loads. The proposed architecture includes four cooperative agents namely DER, ESS, load, and grid agents, each assigned clearly defined responsibilities and coordinated interactions to ensure distributed decision making.

The DC microgrid and the proposed control framework are modeled in MATLAB/Simulink and evaluated under eight distinct operating scenarios representing different generation, load, and grid conditions. Simulation results demonstrate that the proposed approach maintains DC bus voltage stability, ensures reliable supply to critical loads, and improves overall power quality under dynamic operating conditions.

In addition, the system performance is analyzed under communication failures and measurement uncertainties to assess robustness and fault tolerance.

The main contributions of this study are summarized as follows:

- A multi-agent-based energy management architecture specifically designed for DC microgrids that overcomes the flexibility, scalability, and fault tolerance limitations of conventional centralized and single agent approaches.
- A structured coordination mechanism among DER, ESS, load, and grid agents enabling real time distributed control and adaptive power balancing.
- Comprehensive performance evaluation through multi scenario simulation studies, including analysis of fault tolerance under communication disturbances and measurement errors.
- The multi agent framework enables distributed intelligence, enhances resilience against component or communication failures, and provides scalable control suitable for modern DC microgrid applications.

The remainder of this paper is organized as follows. Section II reviews multi-agent-based energy management systems. Section III presents the proposed DC microgrid architecture and agent framework. Section IV discusses the test scenarios and simulation results. Section V concludes the paper.

II. MULTI-AGENT BASED ENERGY MANAGEMENT SYSTEMS

Software agents and the multi-agent systems they form have emerged as powerful technologies for addressing the challenges encountered in implementing information and communication technology applications across diverse domains [16]. Multi-agent systems can be viewed as a collection of autonomous and intelligent entities, called agents, that essentially evolve in an environment that they can perceive and act upon. This environment can be considered anything but the agent itself. These agents have an intelligence of their own that gives them a certain degree of autonomy depending on how they are configured. There are some features that distinguish an agent from classical hardware or software systems. These features, which an agent must have, are presented in Table 2 [17].

In energy management applications, a multi agent system consists of multiple autonomous agents interacting within a shared environment. As illustrated in Figure 1, each agent is structured into three core components: perception, reasoning, and action. The perception module acquires relevant system information such as power levels, voltage measurements, and load status. The reasoning module processes this information using predefined decision rules to determine appropriate control actions. The action module then executes these decisions by adjusting power references, switching states, or coordination signals. Agents communicate through defined communication channels to exchange state information and coordination signals. This distributed

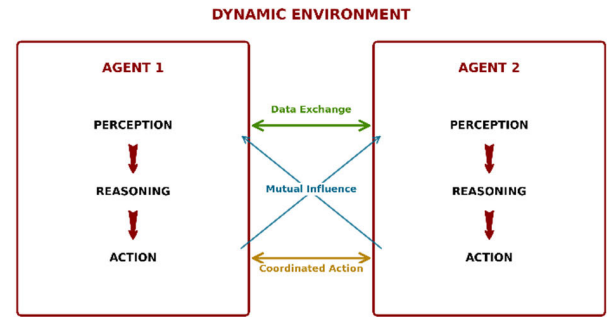


FIGURE 1. Multi-agent systems.

TABLE 2. Properties of agents.

Feature	Description
Autonomy	An agent must be capable of initiating tasks and running without the direct involvement of human users, other agents, or systems.
Reactivity	An agent must constantly perceive its environment and be able to change its knowledge, goals, and actions according to changes in the environment.
Goal-orientedness	An agent must make plans to fulfill what is expected and take the actions required by these plans.
Sociability	An agent interacts with other agents or human users to complete tasks related to their plan.
Permanent continuity	An agent must remain active after completing the tasks it initiated and remain ready to perceive its environment and take the necessary actions.
Mobility	An agent can move from one place to another on the network to fulfill requests, such as performing a transaction on a computer with a more powerful processor and returning or gathering information by browsing information sources on the network.
Learning	An agent can adapt itself to the environment, learn user tendencies and change its actions accordingly.
Rationality	An agent plans for her own goals but does not make plans that would prevent her from achieving their goals.
Veracity	An agent does not knowingly communicate false information.
Benevolence	An agent works to fulfill expectations if it does not conflict with their goals.

interaction architecture enables cooperative decision-making in the absence of centralized supervision. The environment is modeled as the DC microgrid, where variations in generation, storage state-of-charge, and load demand are continuously observed by the agents. Through this perception reasoning action loop and information exchange, the multi agent system achieves coordinated and adaptive energy management. Multi-agent systems are applied to various types of problems in electrical engineering, such as diagnostics, distributed control, modeling and simulation, protection, maintenance planning, etc.

Multi-agent systems have many advantages over classical analytical control methods [18], [19]. The flexible communication structure of multi-agent systems is one of these advantages. Although agents in microgrids only need to have information about their immediate neighbors, they need to communicate with each other to cooperate [20].

This stands out as a decisive reason for choosing multi-agent systems for microgrid energy management. For security reasons, agent communication can be restricted to the local microgrid domain. This localization significantly reduces both the volume of transmitted data and the associated communication costs compared with other communication schemes. In addition, recent studies have highlighted that the increasing connectivity and data exchange in multi-agent-based microgrids introduces new cybersecurity challenges, such as the risk of unauthorized access, data manipulation, or denial-of-service attacks. Therefore, while this study does not provide a detailed cybersecurity analysis, it is important to recognize that implementing secure communication protocols and robust security measures is essential for the reliable and resilient operation of multi-agent-based energy management systems. This remains an important direction for future research.

The combined use of many DERs leads to the design of energy management systems in microgrids. Energy management in microgrids covers a wide range of tasks, including demand response, load forecasting, renewable energy integration and grid optimization [21]. Energy management, in essence, aims to balance supply and demand, maximize efficiency, and minimize costs while ensuring grid stability and resilience. In traditional centralized systems, a single organization typically oversees energy management tasks. However, the decentralized nature of microgrids requires distributed control strategies and energy management systems that can coordinate various energy sources and respond to dynamic operating conditions [22]. In addition to the advantages of distributed control and flexibility, the reliability of the communication infrastructure between agents is a critical aspect in multi-agent-based microgrid systems. Possible malfunctions or measurement errors in the communication network can affect system performance and stability. In this study, the impact of such failures and inconsistent or erroneous data exchange between agents is discussed, and the robustness and fault tolerance of the proposed system under these conditions are briefly assessed.

Multi-agent systems provide a framework for distributed decision-making and coordination among autonomous entities or agents within a microgrid. Each agent represents a component or subsystem within the microgrid, such as generators, ESSs, and loads, equipped with sensing, communication, and decision-making capabilities. Through local interactions and information exchange, agents collaborate to achieve common goals, such as optimizing energy production, minimizing costs, and maintaining grid stability [18].

The versatility of multi-agent systems enables a wide range of energy management applications in microgrids:

- **Distributed Control and Optimization:** Intermediaries in the microgrid collaborate to optimize energy generation, storage, and distribution based on real-time demand and supply conditions. Leveraging local knowledge and preferences, agents can dynamically adjust operating parameters to maximize system

efficiency and reliability. Multi-agent system-based energy management systems have been used in many studies for distributed control and optimization in microgrids [23], [24].

- **Demand Response and Load Management:** Agents interact with end users to implement demand response strategies such as load shedding or shifting to alleviate grid congestion and balance supply and demand. By encouraging consumers to adjust their energy consumption patterns, microgrids can reduce peak demand, alleviate grid instability, and reduce overall energy costs. Multi-agent system-based energy management systems have been used in many studies for this purpose in microgrids [25], [26], [27], [28].
- **Renewable Energy Integration:** Agents coordinate the integration of renewable energy resources such as PV modules and wind turbines into the microgrid. By forecasting renewable energy production and optimizing energy storage and distribution, intermediaries ensure efficient use of renewable resources while maintaining grid stability and reliability. Multi-agent system-based energy management systems have been used in many studies for renewable energy integration in microgrids [20], [29], [30], [31].
- **Fault Detection and Self-Healing:** Agents monitor system health and detect anomalies or errors in real time. In the event of a disruption or failure, agents collaborate to reconfigure the microgrid topology, reroute power flows, and autonomously restore system operation, minimizing downtime and reducing the impact of outages. Multi-agent system-based energy management systems have been used in many studies for fault detection and self-healing in microgrids [32], [33], [34], [35], [36].

III. PROPOSED DC MICROGRID SYSTEM AND AGENTS

In this study, a DC microgrid model comprising a solar PV system, wind turbine, synchronous generator, battery-based ESS, critical and non-critical loads, and grid connection is considered. To ensure coordinated control and effective energy management, four agents are defined in the proposed framework: a DER agent supervising the wind turbine, PV system, and synchronous generator; an ESS agent managing the battery system; a load agent responsible for load coordination; and a grid agent handling power exchange with the utility grid. The system is modeled and simulated in MATLAB/Simulink, and the results are evaluated under various operating conditions. Figure 2 illustrates the proposed DC microgrid configuration and the corresponding agent structure. Each physical component is associated with a dedicated agent, and information exchange among agents enables coordinated decision making to maintain DC bus stability and overall energy balance. The operational logic of each agent is structured into sequential tasks, as detailed in Tables 3-6, where the 'Task Sequence No.' column defines the execution order of these tasks. Further details are provided in the subsequent sections.

A. DER AND DER AGENT

Wind turbine, solar PV modules and synchronous generator constitute DERs. DER agent organizes DERs in the microgrid.

1) WIND TURBINE MODEL

The designed wind energy system consists of a 6 kW three-phase salient-pole permanent magnet synchronous generator (PMSG), a wind turbine, and a pitch angle control system. The simulation parameters are defined with a base power of 6 kW, a base wind speed of 12 m/s, a maximum power at base wind speed of 0.8 p.u., and a base rotational speed of 1 p.u. While the PMSG produces three-phase alternating current, the output voltage and frequency vary according to the wind speed. In the proposed model, a variable wind speed profile ranging from 0 to 12 m/s is applied at specific time intervals. To interface the wind turbine with the common DC bus, a rectifier circuit is employed along with a Maximum Power Point Tracking (MPPT) algorithm to ensure optimal energy conversion efficiency.

2) PV SYSTEM MODEL

The solar PV system model consists of 25 serially connected Suntech STP270S-24 modules. Block parameters in the MATLAB/Simulink model of the created PV system are as follows: open circuit voltage is 44.49V, open circuit voltage is 8.19A, voltage at maximum power point is 35V, current at maximum power point is 7.71A, maximum power value is 270W, temperature coefficient of open circuit voltage is determined as 0.1504% V/deg.C.

3) SYNCHRONOUS GENERATOR MODEL

In order to take precautions against natural variables and sudden output power changes occurring in the output powers of the wind turbine and PV system, the synchronous generator is used to ensure stability in the DC microgrid. Designed synchronous generator model parameters are as follows: nominal power is determined as 60kVA, line-to-line voltage is 800V, frequency is 50Hz, stator resistance is 0.0378pu.

4) DER AGENT

It represents the distributed generation units such as the synchronous generator, wind turbine and PV system in a microgrid. The voltage and current information received separately from these sources are used by this agent. It is also responsible for the MPPT action for wind turbine and PV system. It controls the total power produced in PV system, wind turbine and synchronous generator, and shares it with other agents. The duties and collaborations of the DER agent in the designed microgrid system are presented in Table 3.

B. ESS AND ESS AGENT

The ESS is connected to the common DC bus via a bi-directional DC-DC converter to balance the power consumption and production and to reduce power fluctuations

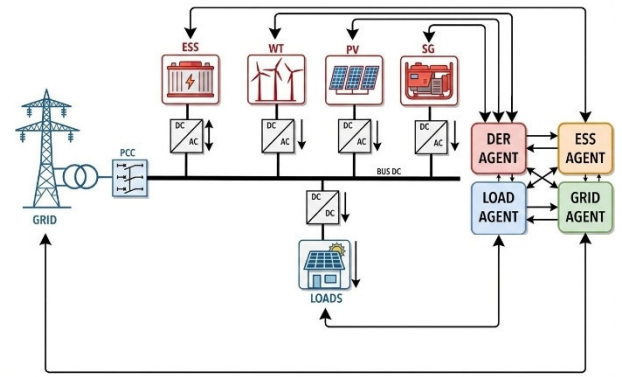


FIGURE 2. The proposed DC microgrid system and agents of the proposed control approach.

TABLE 3. DER agent tasks in DC microgrid.

Task Sequence No.	Tasks
1	The distributed generation agent receives power, voltage, and current information from DERs.
2	The distributed generation agent performs MPPT with the information it receives and transfers it to the resources.
3	The distributed generation agent transfers the generated power information to the grid agent.
4	The distributed generation agent gives production information to the load agent and requests load shedding in case of low production.
5	The distributed generation agent transfers the generated power information to the ESS agent.
6	The difference between distributed generation and load consumption receives its signal from the load agent.
7	It evaluates the signal and decides on energy supply or consumption in the microgrid.
8	Depending on the required action (supply/consumption), it sends requests to the relevant agents following their services.

in the DER output power. A 650V, 10Ah Li-Ion battery is used in the system. ESS agent represents ESS in the microgrid. It controls battery and manages battery charging and discharging, thanks to the production and consumption data, receives from other agents. It also shares battery power information (charging/discharging) with other agents. It monitors the state of charge (SoC) level and requests power from the DER agent and the grid agent when the SoC level is low. When the SoC level drops below the critical level of 20%, it puts the battery into charging mode and transmits information to the DER agent to activate the synchronous generator (if other sources are not available). The duties and collaborations of the ESS agent in the designed microgrid system are presented in Table 4.

C. LOADS AND LOAD AGENT

It represents the loads in the microgrid. It receives power consumption information from all loads in the system and transfers it to other agents. It has the ability to monitor, control and negotiate the power level and connection status of the controllable load. Especially, when the microgrid

TABLE 4. ESS agent tasks in DC microgrid.

Task Sequence No.	Tasks
1	The ESS agent receives voltage, current, power and SoC information from the battery.
2	The ESS agent implements the battery control algorithm and controls the ESS unit.
3	The ESS agent reports the common DC bus voltage information to the grid agent.
4	The ESS agent provides production information to the load agent by requesting that non-critical loads be removed from the system when the SoC is low.
5	The ESS agent provides SoC level information to the DER agent.
6	It independently makes the decision regarding the common DC bus voltage control (operating in charging mode or discharging mode)
7	The ESS agent notifies the DER agent to activate the synchronous generator when the SoC is lower than 20%.

TABLE 5. Load agent tasks in DC microgrid.

Task Sequence No.	Tasks
1	The load agent receives power consumption information from critical and non-critical loads.
2	The load agent controls critical and non-critical loads by looking at the supply/demand status of the system according to the SoC value received from the ESS agent and the production information received from the DER agent.
3	The load agent requests the grid agent to turn on/off the grid connection at the point of common coupling.
4	The load agent requests power from the DER agent.
5	The load agent requests power from the ESS agent.
6	It obtains the total production information from the DER agent and the information of difference between production and consumption. It evaluates this information and ensures that non-critical loads are included in the system again.

is in island (off-grid) mode, it shuts down some non-critical loads depending on the available power. It can distinguish between critical load and non-critical load. It also plays a critical role in the system by sending information to other agents whether the supply-demand balance is achieved or not. The duties and collaborations of the load agent in the designed microgrid system are presented in Table 5.

D. GRID AND GRID AGENT

It represents the grid side within the microgrid. It monitors the grid voltage, phase angle, and frequency, and is responsible for notifying other agents of changes in the microgrid state. It provides control of the power flow between the main grid and microgrid. It controls the connection between the main grid and microgrid. The microgrid is responsible for monitoring and negotiating power from DERs and importing or exporting power when it is operating in grid-connected (on-grid) mode. The duties and collaborations of the grid agent in the designed microgrid system are presented in Table 6.

TABLE 6. Grid agent tasks in dc microgrid.

Task Sequence No.	Tasks
1	The grid agent receives voltage, current and power information from the grid.
2	The grid agent controls the power electronic interface between the common DC bus and AC grid and power flow.
3	The grid agent informs the load agent about the mode (on/off grid).
4	The grid agent reports the operation mode to the distributed generation agent (on/off grid mode).
5	The grid agent reports the operation mode to the ESS agent (on/off grid mode).
6	The grid agent provides operation mode change according to the production information it receives from the DER agent.
7	The grid agent provides operation mode change according to the SoC information it receives from the ESS agent.

E. PROPOSED MULTI-AGENT BASED MICROGRID ENERGY MANAGEMENT SYSTEM

Electrical grids are becoming more complex as the number of DER units in power systems increases. In a microgrid, the central control may not be the feasible method to control many DER units, loads and storage units, because of increasing communication requirements. In the designed DC microgrid, the energy management action is carried out with the multi-agent-based control approach which has inherently distributed control feature to overcome the problems caused by the diversity in production and load sources. In addition to maintaining power quality and supply-demand balance through an effective microgrid management and control framework, the agents also perform battery charge-discharge management, battery lifetime optimization, and common DC bus voltage regulation.

As mentioned earlier, the designed multi-agent system consists of an ESS agent, a DER agent, a load agent, and a grid agent. The design objectives, task allocation, and interaction mechanisms of these agents are detailed in Sections III-A–III-D. The algorithm that governs coordinated energy management, supply-demand balancing, critical and non-critical load control, and operational mode transitions is presented in Figure 3.

The algorithm depicted in Figure 3 illustrates the decision-making process of the proposed multi-agent-based energy management system. Initially, the system determines its operational mode, either grid-connected or islanded (on-grid or off-grid), and communicates this status to the DER and load agents. Based on real time measurements, the system evaluates the balance between power generation and demand. If generation exceeds demand, the excess energy is directed to the ESS agent to charge the battery. Conversely, if demand exceeds generation, the battery is discharged to support the load. When the battery SoC level falls below a predefined threshold, for example 20%, the DER agent activates the synchronous generator to maintain system stability. If power remains insufficient, the load agent sheds non-critical loads to preserve reliable operation. Throughout the process, the grid

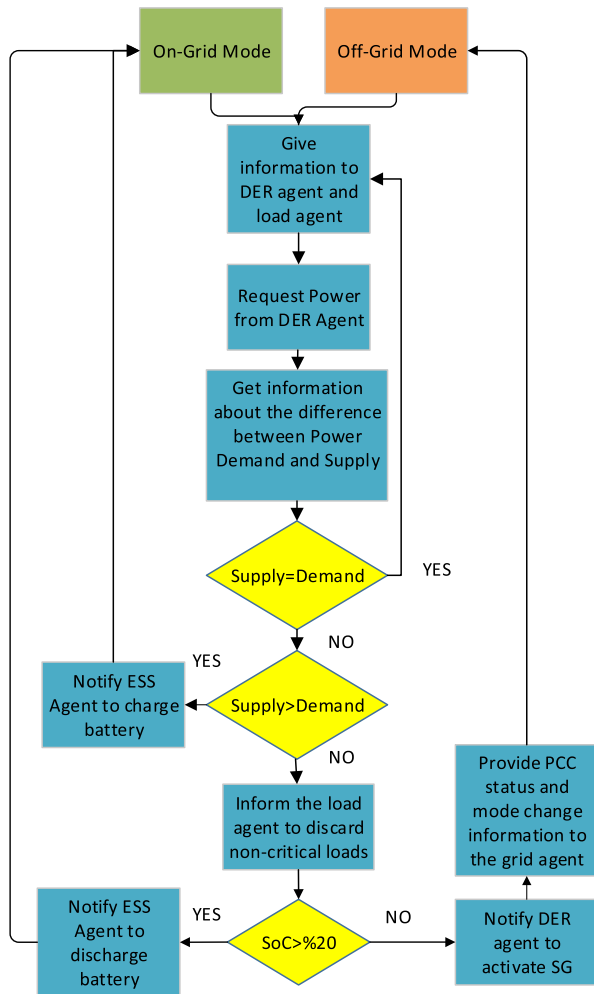


FIGURE 3. The proposed algorithm for energy management.

agent is continuously informed of operational mode and grid connection status to ensure coordinated and seamless system control.

The proposed energy management system employs a rule-based decision method due to its simplicity, transparency, and ease of implementation. By using predefined thresholds and logical rules, the system can respond rapidly to changing operating conditions without relying on computationally intensive optimization algorithms. This enhances real-time performance and improves fault tolerance in distributed DC microgrid environments. In addition, the modular structure of rule-based control supports scalability, as new agents or distributed energy resources can be incorporated without significantly increasing computational complexity. Compared to purely data-driven or heuristic approaches, the proposed method allows easier tuning, adaptation, and practical deployment in heterogeneous microgrid configurations.

In the proposed multi-agent-based architecture, communication and measurement uncertainties are explicitly considered within the decision-making framework. Measurement errors are modeled as Gaussian noise added to sensor signals,

SCENARIO	WT	PV	SG	LOAD	BATTERY	GRID
SCENARIO 1	✓	✓		✓	SOC>%20 DISCHARGE	ON-GRID
SCENARIO 2		✓		✓	SOC>%20 DISCHARGE	ON-GRID
SCENARIO 3	✓	✓	✓	✓	SOC<%20 CHARGE	ON-GRID
SCENARIO 4		✓	✓	✓	SOC<%20 CHARGE	ON-GRID
SCENARIO 5		✓		✓	SOC>%20 DISCHARGE	OFF-GRID
SCENARIO 6				CRITICAL LOAD ONLY	SOC>%20 DISCHARGE	OFF-GRID
SCENARIO 7			✓	CRITICAL LOAD ONLY	OFF	OFF-GRID
SCENARIO 8			OFF	CRITICAL LOAD ONLY	OFF	ON-GRID

FIGURE 4. Scenarios tested with DC microgrid.

while communication imperfections are represented as time varying delays and packet loss events. To enhance robustness, rule-based logic incorporates tolerance margins and threshold bands that prevent unnecessary control actions due to small data perturbations. This mechanism reduces ambiguity in agent responses and improves coordination stability under non-ideal operating conditions.

In the system, power production and consumption are constantly monitored by agents and information is transferred to other agents. When power generation cannot meet the demand, the critical loads continue to be fed, and non-critical loads and grid connection are disconnected, the system continues its operation in off-grid mode. In addition, when the SoC of the battery in the ESS system drops below 20%, the synchronous generator is activated, and the energy management is carried out to provide the necessary power to the system.

IV. TEST SCENARIOS AND SIMULATION RESULTS

The designed DC microgrid was evaluated under eight different operating scenarios, as illustrated in Figure 4. These scenarios were determined with the aim of minimizing the carbon footprint and meeting the power demand of loads within the microgrid as much as possible. Therefore, the control strategy prioritizes meeting demand using renewable energy sources. If these resources are insufficient, the battery-based ESS supplies the deficit. When both renewable sources and the ESS are unable to meet the demand, the synchronous generator is activated. In cases where local generation units are insufficient or unavailable due to operational constraints, the grid provides the required power to maintain system continuity. Additionally, the capacity of each source is also taken into consideration when determining the scenarios. The system ensures the supply of critical loads under all conditions. The battery system consists of a 650V, 10Ah Li-Ion battery connected via a bidirectional DC-DC converter. The minimum SoC threshold is set to 20% to trigger charging or generator activation.

A sampling time of 20 μ s was used in the MATLAB/Simulink simulations. This sampling interval was selected to accurately capture the fast dynamics of the power electronic converters and the DC bus voltage variations while ensuring numerical stability. Sensitivity analysis using smaller sampling intervals confirmed that the system response remains consistent, validating the adequacy of the chosen sampling time.

A. TEST SCENARIOS

In scenario 1, the PV, wind turbine and all loads are activated. Since the battery SoC is $\geq 20\%$, it is discharged. The synchronous generator is disabled, and the system is in grid-connected mode. In this scenario, the battery is constantly discharged because the power produced from renewable resources is lower than the requested power and the battery charge level is $\geq 20\%$. Thus, the battery provides the power required for the system and thus there is no need for power from the synchronous generator. The mathematical relationships between the total supply power (P_s), wind turbine power (P_{wt}), PV system power (P_{pv}), battery output power (P_b), the total load power (P_d), which is the sum of the critical load power (P_{crt_ld}) and non-critical load power ($P_{non-crt_ld}$) are given in Eqs. (1)–(3).

$$P_s = P_{wt} + P_{pv} + P_b \quad (1)$$

$$P_d = P_{crt_ld} + P_{non-crt_ld} \quad (2)$$

$$P_{wt} + P_{pv} < P_d \quad (3)$$

In scenario 2, the PV system and all loads are activated. Battery SoC is $\geq 20\%$ and it is discharging. The synchronous generator and wind turbine are disabled, and the system is in grid-connected mode. Since the generated power is lower than the power demand, the battery is constantly discharged. Although the level of power produced is low, since SoC is $\geq 20\%$, it meets the required power by remaining constantly discharged, and thus the power provided by the synchronous generator is not needed. The mathematical relationships are given in Eqs. (4)–(6).

$$P_s = P_{pv} + P_b \quad (4)$$

$$P_d = P_{crt_ld} + P_{non-crt_ld} \quad (5)$$

$$P_{pv} < P_d \quad (6)$$

In scenario 3, the PV system, wind turbine, synchronous generator and all loads are activated. Battery SoC is $< 20\%$ and it is in charging mode. The system is in grid-connected mode. In this scenario, there is power generated from the wind turbine and PV system. In addition, since the battery SoC is lower than the critical level and it is charging mode, the synchronous generator is enabled. The synchronous generator ensures that the power demand is fulfilled, and the battery is charged. The mathematical relationships are given in Eqs. (7)–(9).

$$P_s = P_{wt} + P_{pv} + P_{sg} \quad (7)$$

$$P_d = P_b + P_{crt_ld} + P_{non-crt_ld} \quad (8)$$

$$P_{wt} + P_{pv} < P_d \quad (9)$$

where P_{sg} stands for synchronous generator power.

In scenario 4, the PV system, synchronous generator and all loads are activated. The wind turbine is offline, battery SoC is $< 20\%$ and it is in charging mode. The system is in grid-connected mode. In this scenario, since only PV system is generating power, the wind turbine offline and the battery SoC is $< 20\%$, the synchronous generator is activated. The synchronous generator provides the power required for the system. The mathematical relationships are given in Eqs. (10)–(12).

$$P_s = P_{pv} + P_{sg} \quad (10)$$

$$P_d = P_b + P_{crt_ld} + P_{non-crt_ld} \quad (11)$$

$$P_{pv} < P_d \quad (12)$$

In scenario 5, the PV system and loads are activated, the battery SoC is $\geq 20\%$ and it is in discharging mode. The wind turbine and synchronous generator are disabled, and the system operates in island mode. Since the power produced in the scenario is lower than the requested power, the battery is constantly discharged, and the battery provides the required power of the system. Although the generated power level is low, the power demand is met by the battery, which remains in discharge mode continuously due to the state of charge (SoC) being $\geq 20\%$. This eliminates the need for a synchronous generator. The mathematical relationships are given in Eq. (13)–(15).

$$P_s = P_{pv} + P_b \quad (13)$$

$$P_d = P_{crt_ld} + P_{non-crt_ld} \quad (14)$$

$$P_{pv} < P_d \quad (15)$$

In scenario 6, the PV system, wind turbine, synchronous generator and loads are disabled. Battery SoC is $\geq 20\%$ and it is discharging. Only critical loads are active, and the system operates in island mode. In this scenario, since the wind turbine and PV system are disabled, there is no production from renewable energy sources. The battery SoC is $\geq 20\%$ and it provides the necessary power only for critical loads. Other loads are disabled due to insufficient production. Since renewable resources are not available, the battery group provides the power required for critical loads as long as the charge level allows. The mathematical relationships are given in Eqs. (16)–(18).

$$P_s = P_b \quad (16)$$

$$P_d = P_{crt_ld} \quad (17)$$

$$(P_{wt} + P_{pv}) = 0 < P_d \quad (18)$$

In scenario 7, the PV system and wind turbine are not available. The battery SoC level is $< 20\%$, and discharge mode is prevented. Besides, since there is not any active renewable source, the battery is not in charging mode (not active). The synchronous generator is active and supplying the critical loads, and the system operates in island mode. The required power for critical loads in the system is provided only by the synchronous generator. The generated power is assumed to be lower than the requested power, therefore, non-critical loads are disabled, and only critical loads are supplied by the generator. The mathematical relationships are given

in Eqs. (19)-(21).

$$P_s = P_{sg} \quad (19)$$

$$P_d = P_{crt_ld} \quad (20)$$

$$(P_{wt} + P_{pv}) = 0 < P_d \quad (21)$$

In scenario 8, the PV system, synchronous generator and wind turbine are disabled. The battery is disabled because its SoC is $<20\%$ and there is no production from renewable energy sources. Only critical loads are active in the system and the system operates in grid-connected mode. In this scenario, the required power for critical loads in the system is provided only from the grid. When the DER agent informs the grid agent that there is no production, the critical load power requested by the load agent is supplied from the grid. P_{grid} refers to power supplied from the grid. The mathematical relationships are given in (22)-(24).

$$P_s = P_{grid} \quad (22)$$

$$P_d = P_{crt_ld} \quad (23)$$

$$(P_{wt} + P_{pv}) = 0 < P_d \quad (24)$$

B. SIMULATION RESULTS

Figure 5 shows the voltage of the common DC bus for all simulated scenarios. The DC bus voltage is consistently maintained at its reference value of 800 V, demonstrating the effectiveness of the multi-agent-based control system. In each scenario, the agents (DER, ESS, load, and grid agents) monitor the system and coordinate actions such as battery charging/discharging, generator activation, or load shedding when needed. For example, if renewable generation is insufficient, the ESS agent discharges the battery, and if the battery SoC drops too low, the DER agent activates the synchronous generator. In cases of power shortage, the load agent prioritizes critical loads. This dynamic and distributed decision-making process among agents ensures that the DC bus voltage remains stable under all tested conditions, as illustrated in Figure 5. As clearly seen in Figure 5, the DC bus voltage remains stable at 800 V throughout all scenarios, with fluctuations not exceeding ± 2 V.

Figs. 6–9 illustrate the voltage, current, SoC, and power profiles of the ESS under different operating scenarios. The battery operating mode depends on the balance between distributed generation and load demand. When the total power generated by the DER units is insufficient to meet the load demand, the ESS agent commands the battery to discharge, resulting in a decrease in SoC. Conversely, when generation exceeds demand, the battery is charged and the SoC increases.

In Scenario 1, shown in Figure 6, the SoC decreases slightly from 60%, indicating battery discharge to support the load under limited renewable generation. The battery voltage remains regulated around 800V, demonstrating stable operation. In Scenario 3 (Figure 7), the SoC increases from approximately 10% as the battery operates in charging mode due to surplus generation and synchronous generator support. Figure 8 presents Scenario 6, where renewable

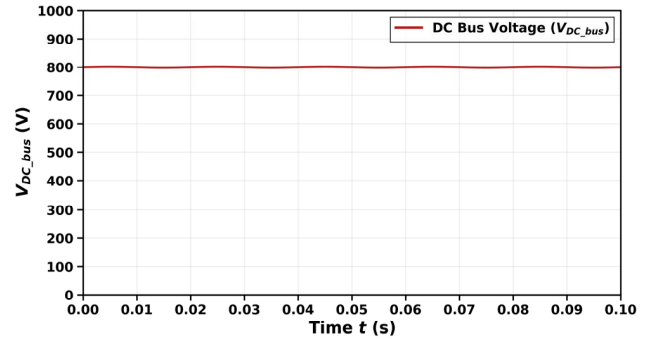


FIGURE 5. Common DC bus voltage graph (All scenarios).

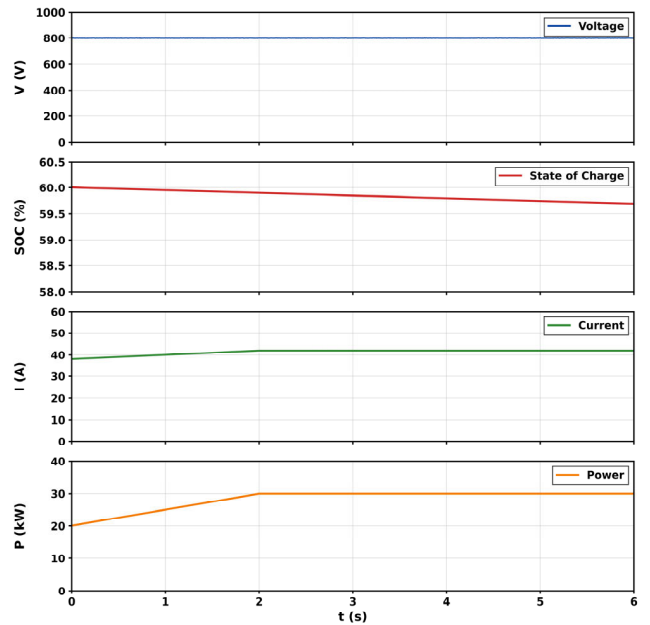


FIGURE 6. Battery voltage, current, charge level and power values for Scenario-1.

sources are unavailable and the system operates in islanded mode. The battery discharges to supply critical loads, with a peak discharge current of approximately 7A. The SoC drops below 30%, highlighting the ESS contribution during constrained operation. In Scenario 8 (Figure 9), the SoC falls below the predefined 20% threshold, resulting in battery deactivation to prevent deep discharge. Under this condition, the grid supplies the required power to maintain system continuity while the battery voltage remains close to the nominal 800 V level.

Figure 10 illustrates the output power profiles of the wind turbine, PV system, synchronous generator, battery system, load, and grid for Scenario 1 under varying wind speed and solar irradiation conditions. The power outputs of the wind turbine and PV system fluctuate in response to environmental variations. As renewable generation changes, the multi-agent-based control framework continuously evaluates the power balance and adjusts the ESS operation accordingly.

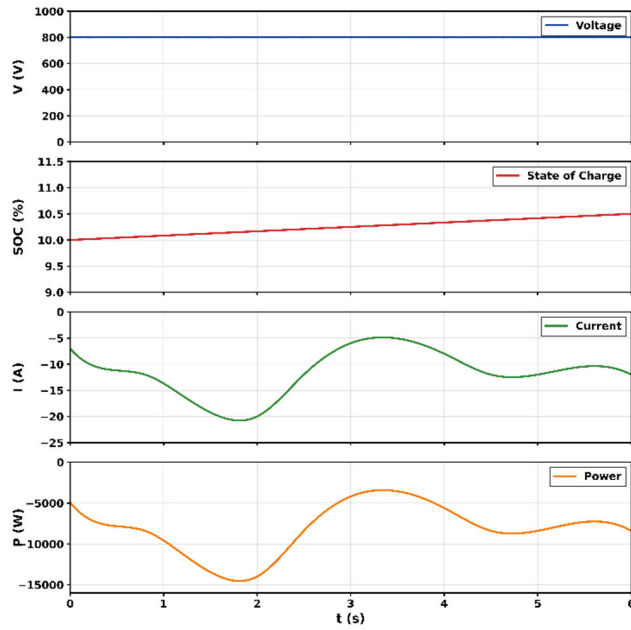


FIGURE 7. Battery voltage, current, charge level and power values for Scenario-3.

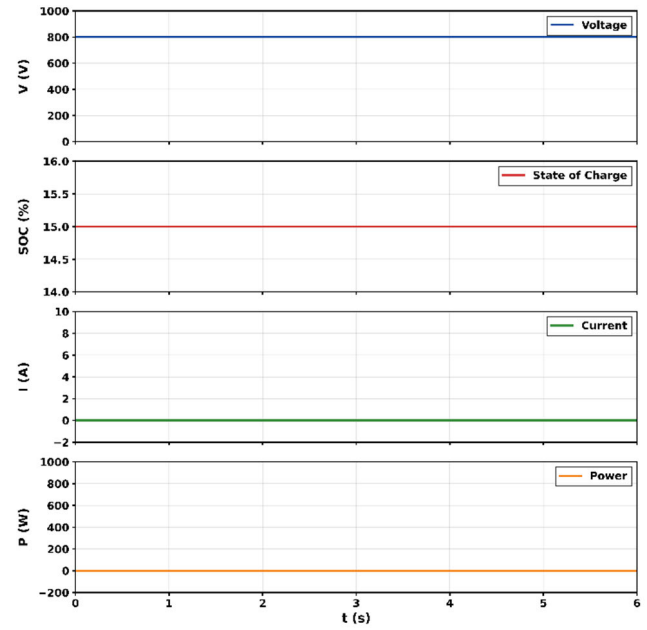


FIGURE 9. Battery voltage, current, charge level and power values for Scenario-8.

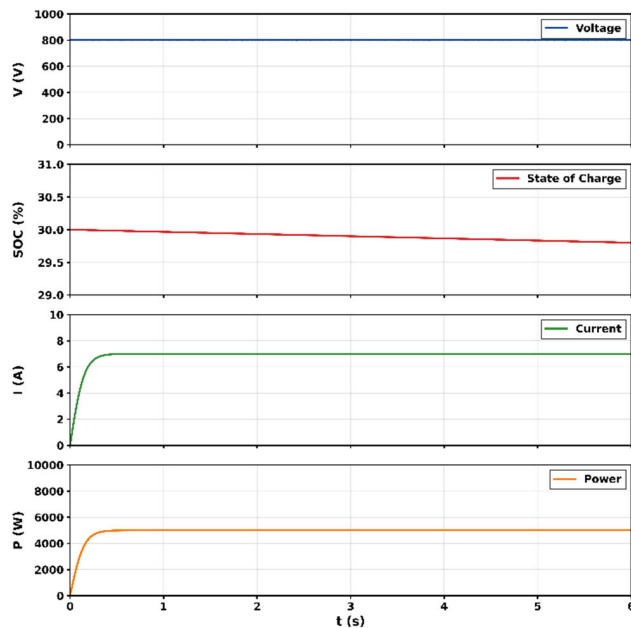


FIGURE 8. Battery voltage, current, charge level and power values for Scenario-6.

In Scenario 1, the total generated power reaches a maximum of approximately 30kW, while the load demand fluctuates around this level. When renewable generation temporarily falls below the load demand, the battery compensates by discharging, as indicated by negative battery power values. This behavior corresponds to the power deficit described in (3). Conversely, during short intervals when renewable output increases, the battery reduces its discharge rate to maintain balance. The synchronous generator remains in standby

or operates at minimal output depending on system requirements, demonstrating coordinated operation among agents. Despite variations in renewable generation and load demand, the DC bus remains stable and supply-demand equilibrium is effectively maintained. The results confirm that the proposed multi-agent-based energy management system dynamically regulates battery power to ensure reliable microgrid operation under fluctuating conditions.

Figure 11 presents the power profiles of the microgrid components in Scenario 3. The outputs of the wind turbine and PV system vary according to wind speed and solar irradiance, initially increasing and then stabilizing. In this scenario, the battery SoC is below 20%, which activates charging mode to restore its energy level. Consequently, the synchronous generator operates as the primary power source. Although renewable generation fluctuates, the synchronous generator maintains a constant output of approximately 10kW to supply the load demand, while the battery absorbs surplus energy and charges up to nearly 15kW. Equation (9) indicates that renewable generation alone is insufficient to meet the demand, prompting synchronous generator activation and coordinated battery charging. The agents determine the battery mode, grid connection status, and load management strategy to ensure energy sustainability and supply-demand balance under variable generation conditions.

Figure 12 illustrates Scenario 4. In this case, the wind turbine is disabled, and its output remains zero. The PV system output varies with solar irradiance and reaches approximately 6kW. Since renewable generation is insufficient to meet the load demand, the synchronous generator supplies the required power with an output of about 20kW. At the same time, the battery operates in charging mode

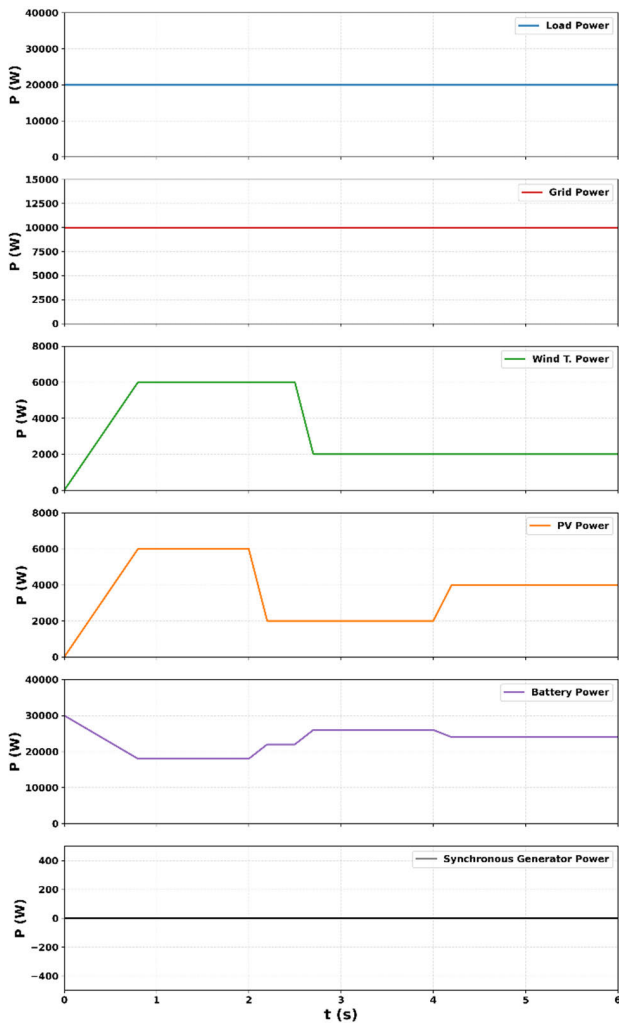


FIGURE 10. Change graph of all power values in the modeled system (Scenario 1).

and absorbs up to 8kW to increase its state of charge. Equation (12) confirms that PV generation alone cannot satisfy the demand, leading to synchronous generator activation and battery charging. The coordinated control strategy ensures stable operation despite changes in renewable availability.

Figure 13 presents the power profiles of the microgrid components in Scenario 6. In this case, both the wind turbine and PV system are inactive, resulting in zero renewable generation. The system operates in island mode, and the grid connection is disabled. Since the battery discharge capacity is insufficient to supply both critical and non-critical loads, the load agent sheds non-critical loads to maintain system stability. The battery SoC remains above the critical threshold; therefore, the synchronous generator is not activated. As seen in the figure, the power demand of the critical loads in the system is met by the battery. The battery supplies up to 5kW to meet the demand of critical loads, continuously discharging to sustain islanded operation.

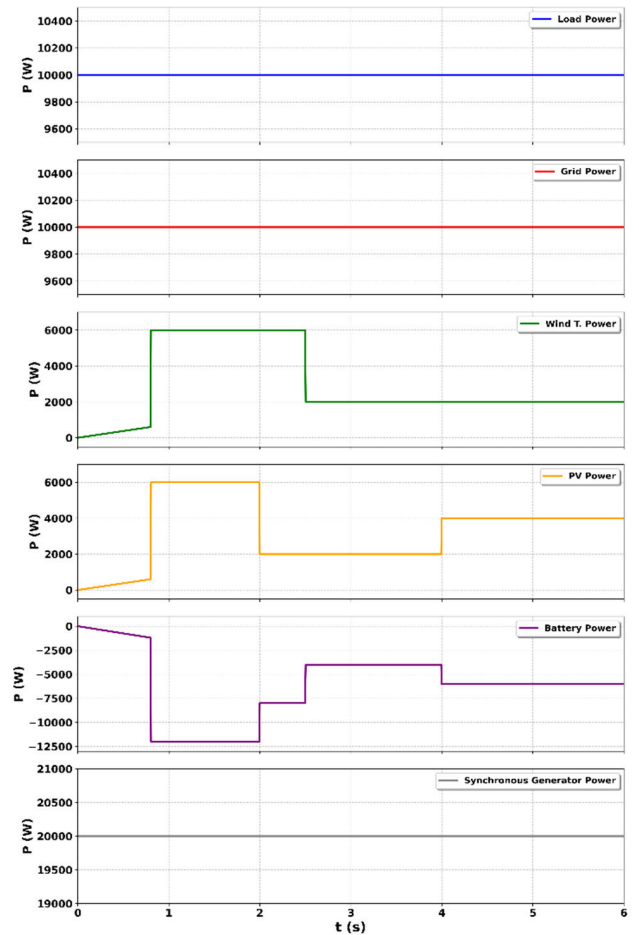


FIGURE 11. Change graph of all power values in the modeled system (Scenario 3).

Equation (18) confirms that renewable generation is zero in this scenario, and the battery exclusively supports critical loads.

Figure 14 illustrates Scenario 7, where renewable sources and the battery are disabled. Under this condition, the synchronous generator operates as the sole power source in island mode. Non-critical loads are shed by the load agent to maintain balance between generation and demand. The synchronous generator provides a constant output of approximately 6kW to supply critical loads. Equation (21) indicates zero renewable generation, confirming that the synchronous generator exclusively feeds critical loads in this scenario.

Figure 15 summarizes the overall production, consumption, and power imbalance across all scenarios. The results demonstrate that critical load demand is satisfied in every case. The control strategy prioritizes critical loads, sheds non-critical loads when necessary, and dynamically switches between grid-connected and islanded modes to maintain operational stability. When local generation and storage are insufficient, the grid supplies the required power. Conversely, when production exceeds consumption, non-critical loads

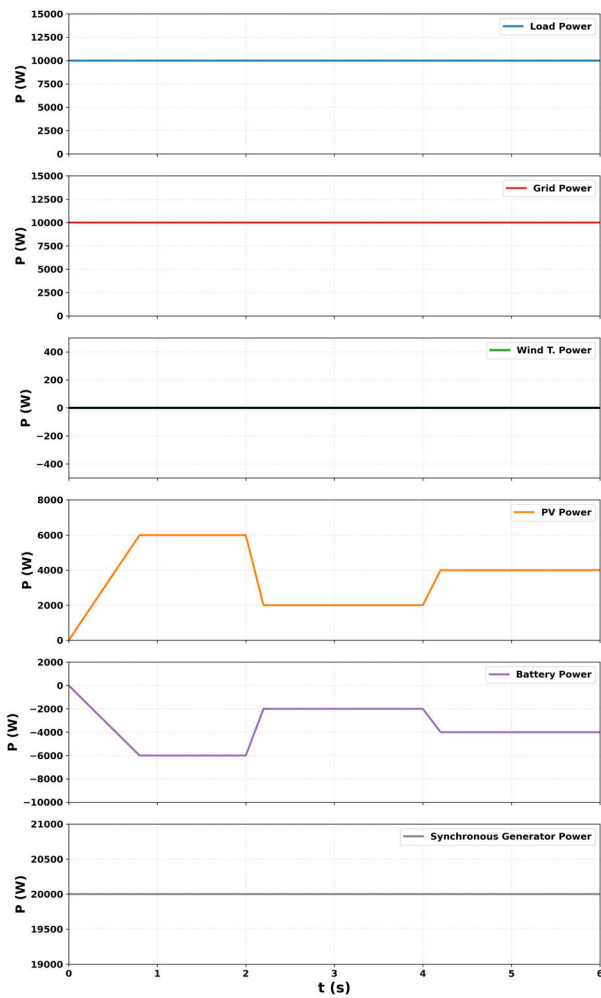


FIGURE 12. Change graph of all power values in the modeled system (Scenario 4).

remain active and grid-connected operation is maintained depending on the available surplus power. Equation (24) describes the operating condition where renewable sources are inactive, and the grid supports the system.

The proposed multi-agent-based energy management system is compared with classical centralized and AI-based energy management approaches. Classical centralized systems typically offer moderate voltage stability and low computational load but lack scalability and fault tolerance. AI-based methods provide high power quality and adaptability but often incur high computational costs. The proposed multi-agent system combines the advantages of distributed control, achieving high voltage stability, robust fault tolerance, and moderate computational load, making it suitable for complex DC microgrid applications.

To evaluate the robustness of the proposed multi agent system, additional simulations were conducted under communication and measurement uncertainty conditions. Communication delays up to 100ms and packet loss rates up to 10% were introduced between ESS and DER agents

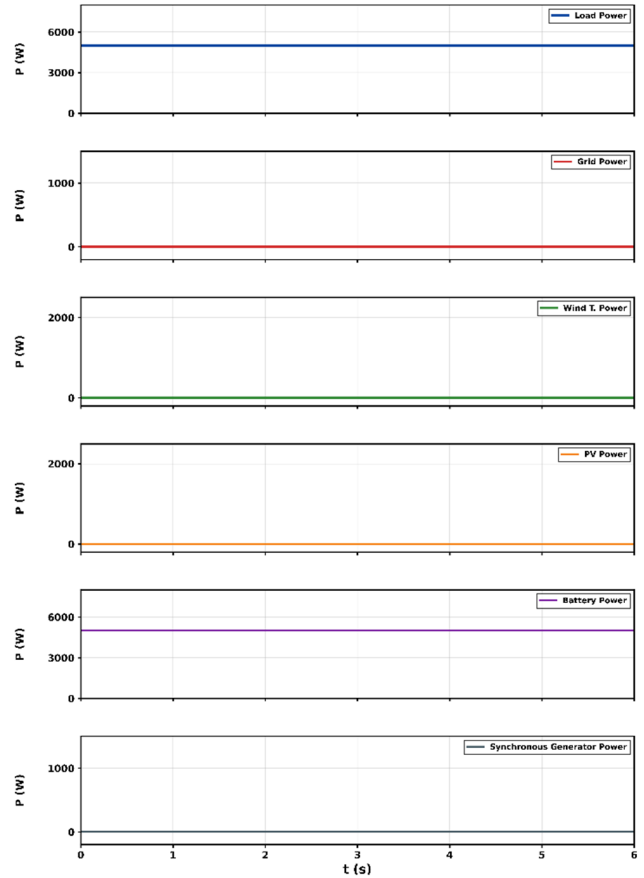


FIGURE 13. Change graph of all power values in the modeled system (Scenario 6).

to assess coordination performance. Measurement errors were modeled as Gaussian noise with a standard deviation of 0.05pu and injected deviations up to 10% in load signals. Under these conditions, the DC bus voltage remained within the $\pm 2\%$ stability margin and critical loads experienced no interruption. Although higher error levels resulted in minor voltage fluctuations and slightly delayed agent responses, overall system stability was preserved. These results demonstrate the inherent fault tolerance of the distributed rule-based coordination mechanism and highlight the importance of reliable communication infrastructure for optimal performance.

The proposed multi-agent-based energy management system is compared with classical centralized and AI-based energy management approaches. Classical centralized systems typically offer moderate voltage stability and low computational load but lack scalability and fault tolerance. AI-based methods provide high power quality and adaptability but often incur high computational costs. The proposed multi-agent system combines the advantages of distributed control, achieving high voltage stability, robust fault tolerance, and moderate computational load, making it suitable for complex DC microgrid applications. This comparison is presented in Table 7.

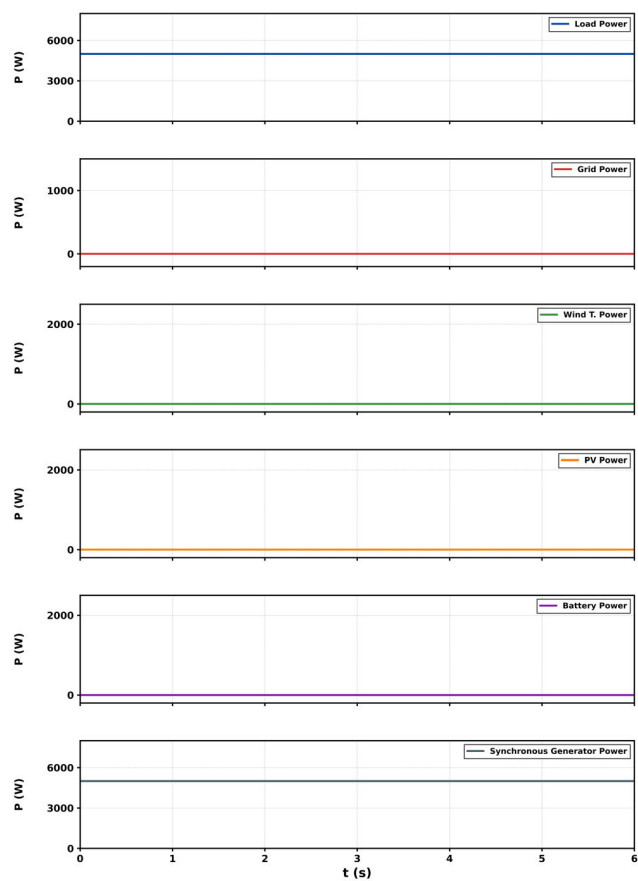


FIGURE 14. Change graph of all power values in the modeled system (Scenario 7).

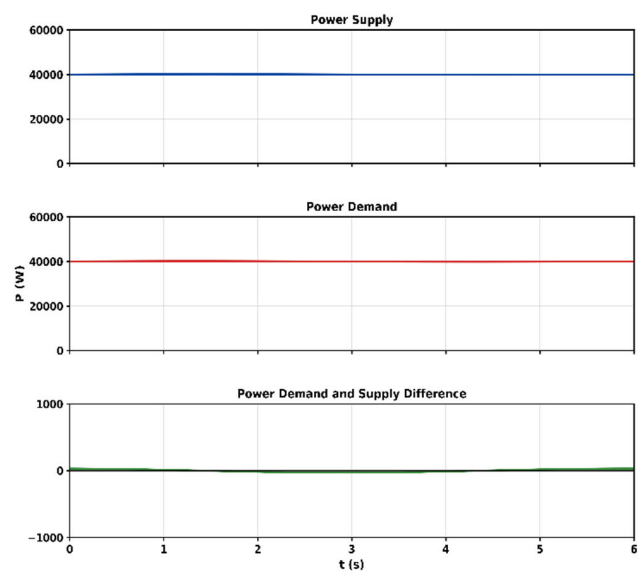


FIGURE 15. Produced and consumed power in the modeled system (All scenarios).

In addition to DC bus voltage stability, several other performance metrics were evaluated to demonstrate the effectiveness of the proposed multi-agent system. These include

TABLE 7. Comparison of energy management methods.

Method	Voltage Stability	Power Quality	Computational Load	Fault Tolerance
Classical Centralized EMS	Moderate	Moderate	Low	Low
AI-Based EMS	High	High	High	Moderate
Proposed Multi-Agent-Based EMS	High	High	Moderate	High

the Power Balancing Accuracy (PBA), which remained above 98.5% across all scenarios, and the Energy Sustainability Index (ESI) for critical loads, which achieved a perfect score of 1.0, indicating zero interruptions. Furthermore, the system’s dynamic response was quantified by the settling time of the battery current and the recovery time of the DER power output during mode transitions. The results show that the proposed MAS architecture ensures not only voltage regulation but also efficient power distribution and high reliability for critical infrastructure.

In addition to voltage stability and power quality, several operational performance indicators were evaluated to assess the practical effectiveness of the proposed system. These include unsupplied energy to non-critical loads during shedding events, the number of load shedding occurrences, the frequency of synchronous generator starts, and the battery cycling count. Across all scenarios, the system maintained low unsupplied energy levels, minimized load shedding events, and optimized generator usage and battery cycling, demonstrating its capability to ensure reliable and efficient microgrid operation.

While practical implementation of multi agent energy management systems in large scale DC microgrids may introduce challenges related to coordination, communication latency, and real time responsiveness, the proposed framework is designed to mitigate these concerns. Unlike optimization-based approaches that require significant computational resources, the proposed architecture is rule based and employs lightweight logical decision mechanisms. Each agent processes locally measured variables and executes predefined control rules, resulting in low computational overhead. Furthermore, the distributed structure eliminates centralized processing bottlenecks and enables parallel execution among agents. This decentralized design enhances scalability and supports real time operation, making the proposed approach suitable for practical deployment in larger microgrid configurations.

Beyond computational and coordination aspects, practical deployment must also consider software complexity, hardware interoperability, and cybersecurity risks. Integration across heterogeneous hardware platforms may introduce compatibility challenges, and synchronization inconsistencies could affect reliability. Furthermore, cybersecurity threats such as unauthorized access, data manipulation, or denial of service attacks may compromise stable operation.

Addressing these issues requires standardized communication protocols, secure authentication mechanisms, encrypted data exchange, and robust system design practices. Careful consideration of these factors is essential to ensure scalable, reliable, and secure implementation of multi-agent-based microgrid control systems in real world applications.

V. CONCLUSION

In this study, an overview of energy management systems for microgrids was presented, and a multi-agent-based energy management approach for DC microgrids was proposed. The framework employs DER, ESS, load, and grid agents to coordinate system operation and maintain power balance. A DC microgrid consisting of a PV system, wind turbine, synchronous generator, battery-based ESS, grid connection, and critical and non-critical loads was modeled in MATLAB/Simulink to evaluate the proposed control architecture under eight distinct operating scenarios. Simulation results verified the effectiveness of the proposed multi-agent controller in maintaining DC bus voltage stability and ensuring reliable power quality under varying operating conditions. The DC bus voltage remained regulated at its reference value across all scenarios. Based on the available generation capacity of DER units, the control framework dynamically adjusted battery operating mode and power level, synchronous generator output, non-critical load status, and grid connection mode. As a result, critical loads were continuously supplied, demonstrating coordinated and resilient system operation.

The proposed distributed and rule-based control strategy provides a scalable and effective solution for DC microgrid energy management. However, several practical limitations remain. Although the rule-based structure reduces computational burden compared to optimization-based approaches, scalability to very large microgrid configurations may introduce coordination and communication challenges that require further investigation. In addition, software integration complexity, hardware interoperability across heterogeneous platforms, and cybersecurity risks from malicious external actors may affect real world deployment. Future work will focus on experimental validation, real time implementation, standardized communication protocols, and enhanced security mechanisms to further strengthen practical applicability.

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The statements made herein are solely the responsibility of the authors.

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